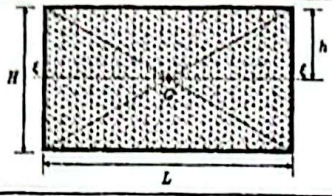
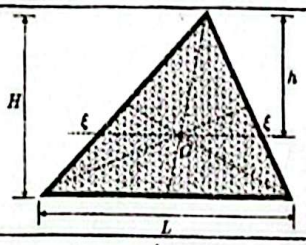
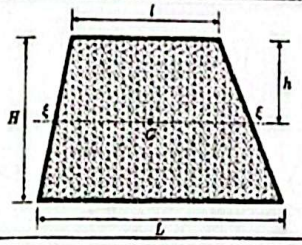
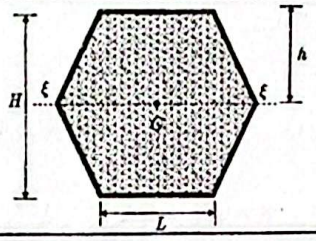
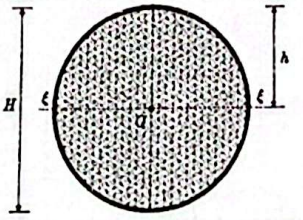
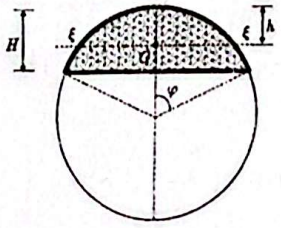
ANNEXE

Table d'unités des barycentres, aires et moment d'inertie

	Forme de la surface	Distance h / Aire S / Moment d'inertie I
Rectangle		$h=\frac{H}{2}$ | $S=LH$ | $I_{\xi\xi}=\frac{LH^3}{12}$
Triangle		$h=\frac{2H}{3}$ | $S=\frac{LH}{2}$ | $I_{\xi\xi}=\frac{LH^3}{36}$
Trapèze		$h=\frac{H}{3}\left(\frac{2L+l}{L+l}\right)$ | $S=\frac{(L+l)H}{2}$ $I_{\xi\xi}=\frac{H^3\left(L^2+4Ll+l^2\right)}{36(L+l)}$
Hexagone régulier		$h=\frac{H}{2}=\frac{a\sqrt{3}}{2}$ | $S=\frac{H^2\sqrt{3}}{2}=\frac{3L^2\sqrt{3}}{2}$ $I_{\xi\xi}=\frac{5H^4}{48\sqrt{3}}=\frac{5L^4\sqrt{3}}{16}$
Cercle		$h=\frac{H}{2}$ | $S=\frac{\pi H^2}{4}$ | $I_{\xi\xi}=\frac{\pi R^4}{4}$
Segment de cercle		$h=R\left[1-\frac{4\sin^3\varphi}{3(2\varphi-\sin 2\varphi)}\right]$ $S=\frac{R^2}{2}(2\varphi-\sin 2\varphi)$ $I_{\xi\xi}=\frac{R^4}{16}\left[4\varphi-\sin 4\varphi-\frac{R^4(1-\cos 2\varphi)^3}{9(2\varphi-\sin 2\varphi)}\right]$